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Rates of Surgical Consultations After Emergency Department Admission in Black and White Medicare Patients

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Supplemental content

IMPORTANCE A surgical consultation is a critical first step in the care of patients with emergency general surgery conditions. It is unknown if Black Medicare patients and White Medicare patients receive surgical consultations at similar rates when they are admitted from the emergency department.

OBJECTIVE To determine whether Black Medicare patients have similar rates of surgical consultations when compared with White Medicare patients after being admitted from the emergency department with an emergency general surgery condition.

DESIGN, SETTING, AND PARTICIPANTS This was a retrospective cohort study that took place at US hospitals with an emergency department and used a computational generalization of inverse propensity score weight to create patient populations with similar covariate distributions. Participants were Medicare patients age 65.5 years or older admitted from the emergency department for an emergency general surgery condition between July 1, 2015, and June 30, 2018. The analysis was performed during February 2022. Patients were classified into 1 of 5 emergency general surgery condition categories based on principal diagnosis codes: colorectal, general abdominal, hepatopancreatobiliary, intestinal obstruction, and upper gastrointestinal.

EXPOSURES Black vs White race.

MAIN OUTCOMES AND MEASURES Receipt of a surgical consultation after admission from the emergency department with an emergency general surgery condition.

RESULTS A total of 1 686 940 patients were included in the study. Of those included, 214 788 patients were Black (12.7%) and 1 472 152 patients were White (87.3%). After standardizing for medical and diagnostic imaging covariates, Black patients had 14% lower odds of receiving a surgical consultation (odds ratio [OR], 0.86; 95% CI, 0.85-0.87) with a risk difference of −3.17 (95% CI, −3.41 to −2.92). After standardizing for socioeconomic covariates, Black patients remained at an 11% lower odds of receiving a surgical consultation compared with similar White patients (OR, 0.89; 95% CI, 0.88-0.90) with a risk difference of −2.49 (95% CI, −2.75 to −2.23). Additionally, when restricting the analysis to Black patients and White patients who were treated in the same hospitals, Black patients had 8% lower odds of receiving a surgical consultation when compared with White patients (OR, 0.92; 95% CI, 0.90-0.93) with a risk difference of −1.82 (95% CI, −2.18 to −1.46).

CONCLUSIONS AND RELEVANCE In this study, Black Medicare patients had lower odds of receiving a surgical consultation after being admitted from the emergency department with an emergency general surgery condition when compared with similar White Medicare patients. These disparities in consultation rates cannot be fully attributed to medical comorbidities, insurance status, socioeconomic factors, or individual hospital-level effects.

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Racial disparities in quality and outcomes of surgical care may be introduced at multiple points along a patient's care trajectory.[1] Among the 5 phases of care described by the American College of Surgeons,[2] the preoperative phases are particularly critical to outcomes for patients who present with general surgery emergencies.[3] For patients with time-sensitive needs, treatment decisions require appropriate triage. Surgical consultation is a crucial first step in the emergency department.

There has been significant work demonstrating unequal access to specialized care for Black patients in the outpatient setting. The main driver of these disparities is thought to be referral bias. Black patients are referred to specialists less often and later in their disease course.[4,5] These trends hold true even among equally insured populations.[6]

Similar patterns of care have been demonstrated in the emergency setting where Black patients have been consistently undertriaged when compared with White patients.[7,8] Undertriage can result in delays in consultation and subsequent treatment. For example, in a single-center study,[9] Black trauma patients had a significantly increased time to consultation compared with White patients. Similarly, in another, single-center study[10] where ophthalmologic care was available around the clock, it was shown that Black patients were 15% less likely than White patients to receive a consultation when presenting for an eye-related condition.

To our knowledge, a nationwide study of surgical consultations has not been performed. Moreover, to date, no one has compared the use of surgical consultations between Black patients and White patients admitted from the emergency department where similar access to urgent care is expected. Therefore, we conducted a nationwide retrospective cohort study aimed at comparing the rates of surgical consultations between Black patients and White patients admitted from the emergency department with an emergency general surgery condition. We hypothesized that Black patients would receive fewer surgical consultations than White patients and that the difference would decrease after standardization.

Key Points

Question Do Black Medicare patients and White Medicare patients receive similar rates of surgical consultations after being admitted from the emergency department with an emergency general surgery condition?

Findings In this cohort study of 1 686 940 patients, after controlling for medical comorbidities, rates of diagnostic imaging, socioeconomic status, and hospital-level factors, Black Medicare patients were significantly less likely to receive a surgical consultation when compared with similar White patients.

Meaning The finding that Black Medicare patients are less likely to receive a surgical consultation than similar White patients could potentially contribute to unfavorable outcomes.

Methods

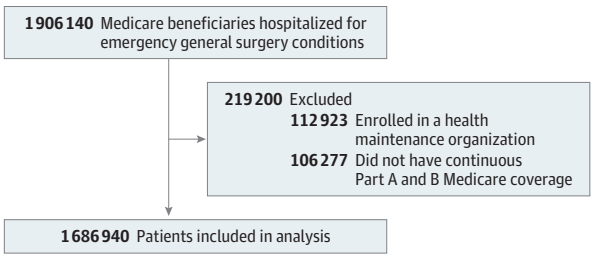
Data Source and Population

The US Centers for Medicare and Medicaid Services Master Beneficiary Summary File, Inpatient, Carrier (Part B), and Durable Medical Equipment files were used for this study. We included patients hospitalized with a principal diagnosis for an emergency general surgery (EGS) condition who were 65.5 years or older, enrolled in Medicare fee-for-service, and presented to the emergency department between July 1, 2015, and June 30, 2018. Details of this coding strategy and the qualifying *Current Procedural Terminology* (*CPT*) codes used are available in the eMethods and eTable 1 in the Supplement. Emergency general surgery conditions were identified using *International Classification of Diseases, Ninth or Tenth Revision, Clinical Modification* codes (*ICD-9/10-CM*). Dates and ages allowed a 6-month look-back period to confirm the index encounter and gather information regarding comorbidities. Patients were classified into 1 of 5 emergency general surgery condition categories based on their principal diagnosis codes: colorectal, general abdominal, hepatopancreatobiliary, intestinal obstruction, and upper gastrointestinal. Diagnoses included in each category are drawn from the work of Shafi et al[11] and are shown in eTable 1 in the Supplement. Patients who did not have continuous Part A & B coverage or who were enrolled in a health maintenance organization at any time between 6 months prior to and 6 months after their index EGS admission were excluded (**Figure**). This study was approved by the institutional review board of the University of Pennsylvania (protocol number 832059), which waived the requirement for participant informed consent. The study was reported according to the STROBE guidelines.

Covariates

In addition to standard covariates such as age, gender, race, and individual Elixhauser comorbidities, we included a total of 57 covariates that might influence the likelihood of calling a surgical consultation (eTable 2 in the Supplement). Among the covariates, a claims-based frailty index (CFI) and angus sepsis scores were calculated for each patient. CFI is a composite score based on a patient's mobility, use of skilled nursing care, and activities of daily living disability. CFI has been validated in Medicare data and has been shown to predict mortality and number of hospital days.[12] We also included dual-eligibility status and Core-Based Statistical Areas (CBSA) as measures of socioeconomic status. Dually eligible beneficiaries are often low-income beneficiaries enrolled in both Medicare and Medicaid.[13]

Figure. Patient Inclusion and Exclusion Flow Diagram



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This designation was used to further control for potential confounding due to economic differences between Black beneficiaries and White beneficiaries that may have influenced the decision to obtain a formal surgical consultation.

CBSAs are geographic areas assigned at the federal level. CBSAs contain 1 or more counties anchored to an urban center. The US Census office uses CBSA areas to describe economically and socially linked regions. CBSA areas can be used to describe differences in urban or rural geographic areas, as well a patient's ease of accessing health care.[14] Because the use of imaging has been shown to differ between Black patients and White patients with abdominal pain[15] and imaging may influence the decision to call a consult, we assigned patient imaging status (abdominal radiograph, chest radiograph, and computed tomography scan) based on the presence of *CPT* codes for each imaging study (eTable 1 in the Supplement). Additionally, we calculated the distance between a patient's home and the hospital. We included travel distance as another covariate to adjust for rural-urban disparities. Travel distance was transformed using the Winsor method to limit the effect of extreme outliers.[16]

Outcomes

The primary outcome of the study was the receipt of a surgical consultation. Surgical consultations were identified by the receipt of an operation during the index hospitalization or by the presence of a qualifying consultation *CPT* code (99221-99223, 99231-99233, 99281-99285, 99218-99220, 99224, 99226, 99217, and 99234-99236) billed by a physician credentialed for general surgery within the first 2 days of presentation to the hospital. We restricted the time frame to within 2 days of presentation to stay consistent with existing studies of emergency general surgery and to capture the consultations that were most likely attributable to the reason for admission. The receipt of a consultation was indicated in our results as C while no consultation was indicated as NC. Receipt of an operation was examined as a secondary outcome. Operations were identified using standard *CPT* codes published in our prior works.[17] A patient with a qualifying operation on hospital day 0, day 1, or day 2 was assigned to the operative treatment group.

Statistical Approach

Descriptive statistics were reported as numbers of patients from each racial group and by consultation status. Continuous variables were reported as mean standard deviation. Univariate analysis of continuous variables was performed using 1-way analysis of variance for normally distributed data and an equality of medians test for nonparametric data. Categorical variables were analyzed using the χ^2 test.

Standardization Using Covariate Balance

We used balancing weights to account for differences in patient characteristics between Black patients and White patients that could contribute to observed differences in consultation rates. Balancing weights are a computational generalization of inverse propensity score weight that we use to create patient populations with similar covariate distributions.[18] This allows us to isolate the association with race by ensuring that the White patients and Black patients we compared were as similar as possible. To achieve this, we set the balance constraints, such that the standardized mean difference in covariates between Black patients and White patients is less than 1%. We estimated 2 sets of weights. The first set of weights ensured that balance held within states. The second set of weights ensured that balance held within hospitals. We then fit generalized linear regression models with logistic links weighted by the balancing weights to estimate outcome differences.

We performed our analysis in a stepwise weighted fashion. This allowed us to delineate how different factors affected the rates of surgical consultation for our cohort. Our first comparison was performed after our population was balanced on all medical and diagnostic imaging covariates. We balanced on diagnostic imaging because it provides addition information on the patients' conditions which may influence the decision to call a consultation with a surgeon. Our second comparison was done with medical and diagnostic imaging covariates, in addition to dual eligibility, CBSA, and travel distance. Then, because individual hospitals often have different rates and thresholds for calling consults,[19] we performed our third stepwise analysis, which limited comparisons to Black covariate-balanced patients and White covariate-balanced patients who were seen in the same hospitals. Statistical significance was based on *P* values less than .05. Data were analyzed using Stata version 17.0 (StataCorp) and RStudio (RStudio Team).

Results

A total of 1 686 940 patients were included in the study. Among the 214 788 Black patients in the study, 61 534 received a surgical consultation (28.6%) and 153 254 did not (71.4%). The median age of Black patients who received a consultation was similar to Black patients who did not receive a consultation (consultation [C], 71.0; 95% CI, 70.0-71.0 vs no consultation [NC], 71.0; 95% CI, 70.0-71.0; $P = .32$). Among the 1 472 152 White patients in the study, 508 725 received a surgical consultation (34.6%) and 963 427 did not (65.4%). The median age of White patients who received a consultation was significantly lower than the mean age of White patients who did not receive a consultation (C, 76.0; 95% CI, 76.0-76.0 vs NC, 77.0, 95% CI, 77.0-77.0; $P < .001$). Patterns of patient characteristics were similar for Black patients and White patients who did and did not receive a consultation for most covariates. Black patients and White patients who received a consultation, on average, had fewer comorbidities, were less likely to have sepsis and had lower frailty scores (eTable 3 in the Supplement).

Most Black patients and White patients in the study had at least 1 comorbidity. Only 3326 of Black patients (1.5%) and 52 425 of White patients (3.6%) had 0 comorbidities. Black patients in the study had higher rates of 3 or more comorbidities compared with White patients. The rates of diabetes, depression, hypertension, obesity, and severe kidney failure had the largest differences between racial groups (**Table 1**). A full

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Table 1. Characteristics of Patients Based on Receipt of a Surgical Consultation Stratified by Race

Characteristic	Consultation No. (%) White	Black	Total	P value	No consultation No. (%) White	Black	Total	P value
Total No.	508 725 (89.2)	61 534 (10.8)	570 259 (100.0)	NA	963 427 (86.3)	153 254 (13.7)	1 116 681 (100.0)	NA
Age, mean (SD), y	75.1 (11.7)	69.6 (14.0)	74.5 (12.0)	<.001	75.9 (12.2)	69.8 (14.3)	75.1 (12.7)	<.001
Gender								
Male	229 852 (45.2)	25 849 (42.0)	255 701 (44.8)	<.001	419 304 (43.5)	64 914 (42.4)	484 218 (43.4)	<.001
Female	278 873 (54.8)	35 685 (58.0)	314 558 (55.2)		544 123 (56.5)	88 340 (57.6)	632 463 (56.6)	
Comorbidities[a]								
Congestive heart failure	82 183 (16.2)	13 308 (21.6)	95 491 (16.7)	<.001	258 896 (26.9)	49 420 (32.2)	308 316 (27.6)	<.001
Controlled diabetes	61 235 (12.0)	13 073 (21.2)	74 308 (13.0)	<.001	163 168 (16.9)	43 303 (28.3)	206 471 (18.5)	<.001
Depression	73 585 (14.5)	6531 (10.6)	80 116 (14.0)	<.001	175 753 (18.2)	19 916 (13.0)	195 669 (17.5)	<.001
Controlled hypertension	140 584 (27.6)	25 622 (41.6)	166 206 (29.1)	<.001	349 284 (36.3)	79 677 (52.0)	428 961 (38.4)	<.001
Obesity	82 442 (16.2)	13 657 (22.2)	96 099 (16.9)	<.001	151 789 (15.8)	32 036 (20.9)	183 825 (16.5)	<.001
Severe kidney failure	24 859 (4.9)	10 187 (16.6)	35 046 (6.1)	<.001	75 920 (7.9)	36 470 (23.8)	112 390 (10.1)	<.001
Hypothyroidism	106 045 (20.8)	6399 (10.4)	112 444 (19.7)	<.001	220 829 (22.9)	17 679 (11.5)	238 508 (21.4)	<.001
Elixhauser Comorbidity Index								
0	29 059 (5.7)	1658 (2.7)	30 717 (5.4)	<.001	23 366 (2.4)	1668 (1.1)	25 034 (2.2)	<.001
1	76 384 (15.0)	6017 (9.8)	82 401 (14.4)		85 911 (8.9)	8104 (5.3)	94 015 (8.4)	
2	98 834 (19.4)	9769 (15.9)	108 603 (19.0)		142 486 (14.8)	16 549 (10.8)	159 035 (14.2)	
3+	304 448 (59.8)	44 090 (71.7)	348 538 (61.1)		711 664 (73.9)	126 933 (82.8)	838 597 (75.1)	
EGS condition								
Colorectal	76 683 (15.1)	9286 (15.1)	85 969 (15.1)	<.001	202 026 (21.0)	29 180 (19.0)	231 206 (20.7)	<.001
General abdominal	66 000 (13.0)	9641 (15.7)	75 641 (13.3)		121 046 (12.6)	25 603 (16.7)	146 649 (13.1)	
HPB	129 586 (25.5)	13 406 (21.8)	142 992 (25.1)		92 780 (9.6)	13 835 (9.0)	106 615 (9.5)	
Intestinal obstruction	119 075 (23.4)	14 082 (22.9)	133 157 (23.4)		36 330 (3.8)	5037 (3.3)	41 367 (3.7)	
Upper GI	100 407 (19.7)	12 408 (20.2)	112 815 (19.8)		295 277 (30.6)	45 947 (30.0)	341 224 (30.6)	
Claims-Based Frailty Index, mean (SD)	0.162 (0.061)	0.174 (0.069)	0.163 (0.062)	<.001	0.183 (0.071)	0.194 (0.077)	0.185 (0.072)	<.001
Angus sepsis (preoperative), yes	61 039 (12.0)	8422 (13.7)	69 461 (12.2)	<.001	156 511 (16.2)	24 803 (16.2)	181 314 (16.2)	.55
Alzheimer/dementia, yes	58 652 (11.5)	8662 (14.1)	67 314 (11.8)	<.001	158 836 (16.5)	28 292 (18.5)	187 128 (16.8)	<.001

Abbreviations: EGS, emergency general surgery; GI, gastrointestinal; HPB, hepatopancreatobiliary; NA, not applicable.

[a] Select representative comorbidities.

list of the individual comorbidities broken down by racial group is shown in eTable 4 in the Supplement.

Black patients had higher rates of dual eligibility when compared with White patients (Black patients, 108 510 of 214 788 [50.6%] vs White patients, 319 916 of 1 472 152 [21.8%]; $P < .001$). Black patients also had lower rates of diagnostic imaging (Black patients, 153 840 of 214 788 [71.6%]; $P < .001$) and had a lower median (IQR) distance between their home and the hospital (Black patients, 4.9 [2.2-10.5] miles; White patients, 6.4 [2.5-14.7] miles). Additionally, Black patients were more likely to be seen at a large (Black patients, 129 410 of 214 454 [60.7%] vs White patients, 679 823 of 1 472 057 [46.4%]; $P < .001$) teaching hospital (Black patients, 179 255 of 214 454 [83.6%] vs White patients, 1 118 574 of 1 472 057 [76.0%]; $P < .001$) located in a metropolitan area (Black patients, 196 080 of 214 454 [91.9%] vs White patients, 1 272 168 of 1 472 057 [86.8%]; $P < .001$) than White patients.

Covariate Balance Results

After balancing, all standardized mean differences for the medical, socioeconomic, and within hospital covariate balance

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Table 2. Odds of Receiving a Surgical Consultation Among Black Patients Compared With White Patients[a]

Characteristic	Risk difference (95% CI)	Odds ratio (95% CI)	
		Unadjusted	Adjusted
Medical			
Black patients	3.17 (−3.41 to −2.92)	0.76 (0.75 to 0.77)	0.86 (0.85 to 0.87)
Socioeconomic status			
Black patients	−2.49 (−2.75 to −2.23)	0.76 (0.75 to 0.77)	0.89 (0.88 to 0.90)
Within hospital			
Black patients	−1.82 (−2.18 to −1.46)	0.78 (0.77 to 0.78)	0.92 (0.90 to 0.93)

[a] White patients used as reference group.

groups were below 1%. Standardized mean differences are shown in eTable 2 in the Supplement.

Unadjusted Outcomes

When compared with White patients, Black patients had 24% lower odds of receiving a surgical consultation (OR, 0.76; 95% CI, 0.75-0.77), with a risk difference of −5.91 (95% CI, −6.12 to −5.69). When we compared patients within the same hospital, Black patients had 22% lower odds of receiving a surgical consultation when compared with White patients (OR, 0.78; 95% CI, 0.77-0.78) with a risk difference of −5.46 (95% CI, −5.68 to −5.24).

Adjusted Outcomes

After standardizing for medical and diagnostic imaging covariates, Black patients had 14% lower odds of receiving a surgical consultation (OR, 0.86; 95% CI, 0.85-0.87) with a risk difference of −3.17 (95% CI, −3.41 to −2.92). After standardizing for socioeconomic covariates, Black patients remained at an 11% lower odds of receiving a surgical consultation compared with similar White patients (OR, 0.89; 95% CI, 0.88-0.90) with a risk difference of −2.49 (95% CI, −2.75 to −2.23). Additionally, when restricting the analysis to Black patients and White patients who were treated in the same hospitals, Black patients had 8% lower odds of receiving a surgical consultation when compared with White patients (OR, 0.92; 95% CI, 0.90-0.93) with a risk difference of −1.82 (95% CI, −2.18 to −1.46) (**Table 2**).

Rates of Operative Treatment

Rates of operative treatment differed significantly by race among those who had a surgical consultation both before and after each sequential standardization (before standardization, Black patients, 23.5%; 95% CI, 23.2-23.9 vs White patients, 30.4%; 95% CI, 30.2-30.5; $P < .001$); after standardization on age, EGS condition, sepsis, CFI, hospitalization 6 months prior, medical comorbidities, and diagnostic imaging: Black patients, 23.5%; 95% CI, 23.2-23.9 vs White patients, 27.5%; 95% CI, 27.4-27.6; $P < .001$; after standardizing on age, EGS condition, sepsis, CFI, hospitalization 6 months prior, medical comorbidities and diagnostic imaging with dual eligibility, distance and rural/urban: Black patients, 23.5%; 95% CI, 23.2-23.9 vs 27.1% White patients, 27.1%; 95% CI, 26.9-27.2; $P < .001$ and additionally after standardization age, EGS condition, sepsis, CFI, hospitalization 6 months prior, medical comorbidities, and diagnostic imaging with dual eligibility, distance and rural/urban within hospitals: Black patients, 23.3%; 95% CI, 23.2-23.9 vs White patients, 29.8%; 95% CI, 29.6-29.9; $P < .001$).

Discussion

This nationwide Medicare cohort study examined the effect of race on the receipt of a surgical consultation in a universally insured population of older adults. The study demonstrates that even after adjusting for medical comorbidities, socioeconomic factors, and individual hospital-level effects, Black patients are significantly less likely to receive a surgical consultation when admitted from the emergency department with an emergency general surgery condition. Our weighted stepwise analytic strategy shows that controlling for a patient's medical comorbidities, receipt of diagnostic imaging, socioeconomic factors, and individual hospital-level effects merely mitigated the disparity seen between Black patients and White patients without leveling the playing field. This indicates that there are likely additional factors that impede a Black patient's access to surgical care.

Emergency general surgery conditions carry a high risk of complications and mortality, especially in the population of patients 65 years and older.[20] Many EGS conditions can be managed operatively or nonoperatively; however, deciding what management strategy to take can be complex. The involvement of a surgical specialist in decision making for patients with emergency general surgery conditions is critical, as surgeons are trained in both operative and nonoperative management. Our study highlights the need for equitable access to surgical care for all racial groups to ensure expert management for surgical conditions.

Health care disparities are often attributed to patient comorbidities, differences in access to care, socioeconomic barriers, and over use of lower quality hospitals. The strength of this study was in the careful control for these variables with our weighting methodology. Imaging is a critical tool in modern medical practice for triage and diagnosis. Receiving an abdominal computed tomography or radiography can provide important information on a patient's condition and prompt an escalation of care. It has been previously demonstrated that Black patients receive lower rates of diagnostic imaging in the emergency setting.[21] Given this, we believed it was essential to control for the rates of imaging between our racial cohorts. Even after adjustment for differential imaging rates, we demonstrated that Black patients received lower rates of consultation.

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A patient's socioeconomic status can have a strong association with a patient's ability to receive preventative care or access high-quality hospitals.[22] When we controlled for socioeconomic status we saw a reduction in disparities; however, Black patients still had a significantly lower rate of surgical consultations. Consultation rates can also be influenced by an individual hospital's culture and resource availability.[23] These hospital-level variations were another important variable to control for in our analysis, and just as with our medical/diagnostic imaging and socioeconomic adjustment, after controlling for hospital-level factors we saw a reduction in the disparity between Black and White consultation rates; however, again the difference remained significant.

Advanced age has been shown to affect referral patterns[24,25]; however, our Black cohort was younger across all groups than our White cohort. Given this pattern, we would have expected to see greater use of consultations for Black patients. However, we observed the opposite pattern of consultation usage. Furthermore, after comparing groups similar on age, a disparity in the use of consultation remained significant. Alternatively, referral bias, which is sometimes attributed to logistic issues such as childcare, transportation, employment issues, or specialist availability,[26-29] might have explained our findings in an elective setting. However, given that our patient cohort was enrolled in the study while in the emergency department, these factors would not have affected our results. In this setting, the specialist is brought to the patient's bedside whereas in the outpatient setting the patient must travel to the specialist.

Our findings suggest the presence of factors other than patient comorbidity, socioeconomic status, or hospital factors are contributing to the observed disparities in surgical consultations between Black patients and White patients. Individual patient-level factors, not available in claims data, may play an important role in explaining the disparity in receipt of surgical care. For example, Black patients have been reported to have a higher rate of refusing surgical care.[30,31] The reasons for refusal are often attributed to cultural beliefs, distrust, and limited access to subspecialty care.[32] Black patients have also been reported to have lower overall trust in physicians.[33] Some of this is attributed to fears of exploitation by the medical profession and knowledge of prior discrimination against Black patients in health care.[34] While the patient might not have been explicitly asked if it was OK to consultation a surgeon, the emergency department physician might allude to the need for an operation thus, giving a patient the opportunity to refuse surgical care and mitigating the decision to call the surgeon. Black patients have long experienced discrimination in the US health care settings, leading to a justified sense of mistrust toward the system. The purpose of highlighting this issue is to bring cultural context and awareness to clinicians regarding these issues and patient concerns. Awareness of these concerns can help physicians to affectively communicate and engage in shared decision making.

Standardized care guidelines have been shown to reduce clinical subjectivity in fields, such as trauma and burn care.[35,36] Standardizing the way emergency department health care professionals decide to consultation surgical services may help to combat some of the disparities observed in this study. However, careful consideration must be put into defining a standardized care pathway, so not to exclude groups of patients who may benefit from surgical consultation.

In other areas, such as surgical oncology,[37] it has been shown that Black patients are less likely to receive an indicated operation than White patients. The explanation for this disparity is not entirely known. For our study, we are interested in testing whether Black Medicare patients and White Medicare patients receive a surgical consultation in an environment where access to consultation should be equal for both groups. To compare rates of use of surgery in a meaningful fashion, we would have to do another set of experiments to control for potential confounding, such a surgeon's preference for operative vs nonoperative treatment. This is outside of the scope of this study and a different question then the one posed here. However, it is interesting to note that the rates of operative treatment differed significantly by race among those who had a surgical consultation even after standardization to control for observed differences between Black and White patients.

Overall, this study highlights that the unequal access to surgical care we observed cannot be fully attributed to medical comorbidities, insurance status, socioeconomic factors, or individual hospital-level effects. As a medical community, we must further investigate the underlying reasons for these persistent disparities and the impact of these disparities on overall surgical outcomes.

Limitations

The strength of this study is its large national sample size, by using a national Medicare database, we increase our generalizability. While our large sample size improves the generalizability of our study findings, it may also cause small differences in our populations to be statistically significant, these differences should be interpreted critically in regard to their clinical significance. Limitations of the study include its retrospective nature. Use of claims databases also carry some inherent limitations as more granular clinical data are unable to be extracted, specifically patients' physiology and the severity of certain comorbidities, which may play a role in explaining the disparities seen in rates of consultation. Our balancing weighted methodology sought to mitigate differences in comorbid conditions between patients but is limited by the data available in the claims database. Timeliness of a consultation is another critically important aspect of emergency general surgery care. Prior work in the trauma surgery literature that has demonstrated that Black patients receive delayed surgical care, potentially leading to worse outcomes.[9] Timing of consultation is outside the scope of this project but is an important avenue for future work.

Conclusions

In this universally insured, covariate balanced population, Black Medicare patients were found to have lower odds of receiving a surgical consulation compared with similar White

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Medicare patients after being admitted from the emergency department with an emergency general surgery condition. Balancing medical, diagnostic imaging, socioeconomic factors, and individual hospital-level effects mitigated the disparity in the receipt of surgical consultations; however, significant differences persisted.

ARTICLE INFORMATION

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Concept and design: Roberts, Rosen, Keele, Syvyk, Reilly, McHugh, Kelz.
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